Siddharth Rout

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Education

Columbia University in the City of New York — New York, NY
B.A. in Mathematics and Computer Science — *Expected Graduation: May 2029*

- GPA: 3.9/4.0 | Data Science Society, Alpha Kappa Psi | 4x Hackathon Winner (2x MLH, Columbia DevFest, Columbia x NYU); IMC Prosperity 4 Top 2%; Bridgewater x Metaculus Forecasting Top 3%; Columbia Pitch Comp 2nd of 80+; Putnam 2025
- Coursework: Data Structures & Algorithms, Computer Science Theory, Advanced Programming in C, Discrete Mathematics, Probability Theory, Modern Algebra, Linear Algebra, Multivariable Calculus, Ordinary Differential Equations

St. Christopher's School — Bahrain
A-Level Diploma — *Jun 2025*

- A*A*A*A* in Math, Further Math, Computer Science, & Physics; Highest National A-Level Scores (Math, Further Math); 6x UKMT Gold, 2x UKMT Olympiad Bronze; Bahrain Bourse TradeQuest Investment Competition Captain (Best Presentations)

Experience

Founding Engineer — Jan 2026 – May 2026
Union Square Banking Corporation — *New York, NY*

- Designed and built a production agent platform in Python as first engineering hire ($6M General Catalyst-backed); deployed autonomous browser-use agents across 15+ heterogeneous portals, running 30+ minute multi-step workflows with deterministic state checkpointing, recovery, and NLP-based parsing of submission guidelines into structured task graphs
- Engineered Python orchestration backend with object-oriented state machines, fault-tolerant execution, and adaptive retry logic; built natural-language information-retrieval layer (LangChain, MCP, vector embeddings) over 10,000+ unstructured documents

Quantitative Systems Engineer Intern — Jan 2026 – May 2026
RAYS Capital — *Hong Kong (Remote)*

- Built distributed Python and AWS data pipelines ingesting 8 GB/day of streaming market data across 18,000 instruments with sub-200ms ingestion-to-store latency, replacing fragmented manual workflows feeding the firm's core analytics infrastructure
- Designed real-time web dashboards (React, FastAPI, WebSockets, REST APIs) with sub-second push latency and per-stakeholder alert routing; adopted firm-wide across 6 portfolio managers, cutting decision turnaround from 4 hours to under 1 hour (75%+)

Co-founder — Jun 2024 – Dec 2025
Gulf Intel — *Bahrain*

- Architected production full-stack platforms (Python, FastAPI, PostgreSQL, Docker, REST APIs) on live infrastructure for 30+ paying SME clients spanning demand forecasting, predictive ordering, and inventory automation, generating $10K+ ARR

Software Engineer Intern — Jul 2024 – Sep 2024
STC — *Bahrain*

- Built end-to-end multilingual Arabic/English/Hindi NLP transcription system (Python, TensorFlow, Swift) for call-center audio, lifting classification accuracy 40% in low-signal environments via MFCC feature extraction and spectral analysis

Projects

Govo | *iOS Food Recommendation App* — 2025

- Co-founded and shipped on the iOS App Store (Swift, Xcode, FastAPI) with 2,000+ users and 40% week-1 retention; built ML pipeline using pretrained text embeddings and cosine similarity over swipe vectors, hitting 92% pairwise accuracy

Columbia Software Solutions (CSS) | *Founding Team* — 2025 – Present

- Serve on the founding team of Columbia's student org shipping free production software to NYC nonprofits, small businesses, and startups; lead engineering, code review, and mentor project leads across 3+ concurrent semester engagements
- Shipped two production systems for Refettorio Harlem (Node.js, Express, React) replacing paper logs: a Twilio WhatsApp logging service and an offline-first tablet check-in app with sub-15-second flows; featured in The New York Times

Research

Machine Learning Research | *Columbia University Data Science Institute* — Oct 2025 – Apr 2026

- Built course-aware AI tutoring backend (Python, FastAPI, PostgreSQL, LLM-based retrieval) under Dr. Nakul Verma; deployed to 200+ students with sub-300ms median latency, serving 10,000+ weekly interactions across multiple CS courses
- Designed backend services for automated problem parsing, student skill-inference modeling, and adaptive task sequencing via a concept-graph representation driven by curriculum-learning and Zone-of-Proximal-Development principles

Source-Free Unsupervised Domain Adaptation | *Oxford University* — Oct 2023 – Sep 2024

- Designed and evaluated source-free unsupervised domain adaptation algorithms in Python and PyTorch under Dr. Pramit Saha on the Office-Home benchmark, lifting target-domain accuracy 8.2% over the SHOT baseline with modular ML components

Technical Skills

Languages: Python, Java, C, C++, JavaScript/TypeScript, Swift, SQL (PostgreSQL, MySQL), HTML/CSS, Bash
Frameworks & Infra: FastAPI, Node.js, Express, React, Next.js, REST APIs, WebSockets, Docker, AWS, Linux, Git, Xcode
ML & Data: PyTorch, TensorFlow, scikit-learn, NumPy, pandas, SciPy, LangChain, MCP, Playwright, distributed data pipelines